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Class: BE Computer

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Assignment No: 2B OCR Letter Recognition using DNN

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Import libraries

import numpy as np

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense, Dropout

from tensorflow.keras.optimizers import RMSprop

from tensorflow.keras.datasets import mnist

import matplotlib.pyplot as plt

Load dataset

(x_train, y_train), (x_test, y_test) = mnist.load_data()

Show sample image

plt.imshow(x_train[0], cmap='gray')

plt.title("Sample Digit")

plt.show()

Print shapes

print("X_train shape", x_train.shape)

print("y_train shape", y_train.shape)

print("X_test shape", x_test.shape)

print("y_test shape", y_test.shape)

Reshape (same as PDF)

x_train = x_train.reshape(60000, 784)

x_test = x_test.reshape(10000, 784)

Convert to float and normalize

x_train = x_train.astype('float32')

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x_test = x_test.astype('float32')
x_train /= 255
x_test /= 255

# Convert labels to one-hot encoding
num_classes = 10
y_train = np.eye(num_classes)[y_train]
y_test = np.eye(num_classes)[y_test]

# -----
# Model Architecture
# -----
model = Sequential()

# Input + Hidden Layers
model.add(Dense(512, activation='relu', input_shape=(784,)))
model.add(Dropout(0.2))
model.add(Dense(512, activation='relu'))
model.add(Dropout(0.2))

# Output Layer
model.add(Dense(num_classes, activation='softmax'))

# Compile model
model.compile(loss='categorical_crossentropy',
              optimizer=RMSprop(),
              metrics=['accuracy'])

# Train model
batch_size = 128
epochs = 20
history = model.fit(x_train, y_train,
                   batch_size=batch_size,
                   epochs=epochs,

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verbose=1,

validation_data=(x_test, y_test))

# Evaluate model

score = model.evaluate(x_test, y_test, verbose=0)

print("Test loss:", score[0])

print("Test accuracy:", score[1])

```

OUTPUT:

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print("Test loss:", score[0])
print("Test accuracy:", score[1])
... Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11498434/11498434 0s 0us/step
Sample Digit
0
5
10
15
20
25
0 5 10 15 20 25
X_train shape: (60000, 28, 28)
y_train shape: (60000,)
X_test shape: (10000, 28, 28)
y_test shape: (10000,)
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an 'input_shape'/'input_dim' argument to a layer. When using Sequential
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/20
469/469 9s 18ms/step - accuracy: 0.9211 - loss: 0.2579 - val_accuracy: 0.9637 - val_loss: 0.1139

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Epoch 3/20
469/469 11s 18ms/step - accuracy: 0.9769 - loss: 0.0747 - val_accuracy: 0.9731 - val_loss: 0.0867
...
Epoch 4/20
469/469 8s 18ms/step - accuracy: 0.9812 - loss: 0.0595 - val_accuracy: 0.9772 - val_loss: 0.0737
Epoch 5/20
469/469 7s 15ms/step - accuracy: 0.9851 - loss: 0.0487 - val_accuracy: 0.9790 - val_loss: 0.0712
Epoch 6/20
469/469 8s 18ms/step - accuracy: 0.9876 - loss: 0.0401 - val_accuracy: 0.9831 - val_loss: 0.0654
Epoch 7/20
469/469 7s 16ms/step - accuracy: 0.9887 - loss: 0.0341 - val_accuracy: 0.9797 - val_loss: 0.0760
Epoch 8/20
469/469 8s 17ms/step - accuracy: 0.9903 - loss: 0.0301 - val_accuracy: 0.9835 - val_loss: 0.0622
Epoch 9/20
469/469 8s 18ms/step - accuracy: 0.9910 - loss: 0.0268 - val_accuracy: 0.9820 - val_loss: 0.0817
Epoch 10/20
469/469 7s 15ms/step - accuracy: 0.9934 - loss: 0.0218 - val_accuracy: 0.9834 - val_loss: 0.0683
Epoch 11/20
469/469 10s 16ms/step - accuracy: 0.9930 - loss: 0.0215 - val_accuracy: 0.9845 - val_loss: 0.0641
Epoch 12/20
469/469 9s 18ms/step - accuracy: 0.9939 - loss: 0.0180 - val_accuracy: 0.9827 - val_loss: 0.0737
Epoch 13/20
469/469 7s 16ms/step - accuracy: 0.9944 - loss: 0.0165 - val_accuracy: 0.9850 - val_loss: 0.0735
Epoch 14/20
469/469 8s 17ms/step - accuracy: 0.9948 - loss: 0.0157 - val_accuracy: 0.9839 - val_loss: 0.0789
Epoch 15/20
469/469 9s 19ms/step - accuracy: 0.9952 - loss: 0.0145 - val_accuracy: 0.9845 - val_loss: 0.0664
Epoch 16/20
469/469 7s 15ms/step - accuracy: 0.9955 - loss: 0.0129 - val_accuracy: 0.9821 - val_loss: 0.0904
Epoch 17/20
469/469 8s 18ms/step - accuracy: 0.9961 - loss: 0.0129 - val_accuracy: 0.9830 - val_loss: 0.0854
Epoch 18/20
469/469 9s 18ms/step - accuracy: 0.9963 - loss: 0.0105 - val_accuracy: 0.9837 - val_loss: 0.0728
Epoch 19/20
469/469 9s 15ms/step - accuracy: 0.9965 - loss: 0.0111 - val_accuracy: 0.9844 - val_loss: 0.0843
Epoch 20/20
469/469 10s 15ms/step - accuracy: 0.9961 - loss: 0.0111 - val_accuracy: 0.9828 - val_loss: 0.0937
Test loss: 0.09365160018205643
Test accuracy: 0.9828000068664551

```